Improving Melanoma Classification Across Diverse Skin Tones Through Skin-Tone-Aware Image Augmentation

Motivation and Related Work

Melanoma is one of the most severe forms of skin cancer, and early diagnosis significantly improves patient outcomes [1]. Recent advances in computational imaging and deep learning have enabled automated classification of skin lesions from clinical and dermoscopic images. Convolutional neural networks (CNNs) have become the dominant approach and have demonstrated diagnostic performance comparable to dermatologists in some studies [1].

Despite these advances, existing skin lesion classifiers often fail to generalize across diverse patient populations. A major limitation identified in recent literature is the lack of skin tone diversity in publicly available dermatology datasets [2]. Darker skin tones are frequently underrepresented, making it difficult to evaluate whether models perform consistently across all populations [2]. As a result, many existing systems exhibit reduced accuracy on skin-of-color (SOC) populations [2], [3].

Recent work has also shown that state-of-the-art dermatology AI models perform worse on diverse clinical image sets containing darker skin tones. Fine-tuning models with more representative datasets has been shown to reduce this performance gap [4]. These findings suggest that improving dataset diversity may be as important as improving the underlying machine learning model itself.

This project is motivated by the need for more equitable dermatology AI systems. Rather than developing a new classifier, this work will investigate whether skin-tone-aware image augmentation can improve melanoma classification performance across diverse demographic or skin-tone subgroups.

Project Overview

This project proposes a web-based computational imaging framework for evaluating the impact of skin-tone-aware image augmentation on melanoma classification. The system will allow users to upload a skin lesion image and receive a prediction from a pretrained melanoma classification model indicating whether the lesion is suspicious for melanoma or likely benign.

Instead of developing a new deep learning architecture, the project will leverage an existing validated model and focus on improving training data diversity through image augmentation. Augmentation techniques will generate realistic variations in skin pigmentation and imaging conditions while preserving lesion characteristics. The system will also incorporate Grad-CAM heatmap visualizations to highlight image regions most influential to the model's prediction, providing model interpretability and lesion localization.

Publicly available dermatology image datasets containing labeled skin lesion images will be used for model training and evaluation [5], [6]. Classification performance will be compared before and after augmentation using accuracy, precision, recall, F1-score, and ROC-AUC. Where dataset metadata permits, results will also be analyzed across demographic or skin-tone subgroups to investigate whether augmentation improves model robustness and reduces potential performance disparities.

The expected outcome is a proof-of-concept web application and an experimental evaluation of whether skin-tone-aware augmentation improves fairness and robustness in melanoma classification systems. By combining image augmentation, classification, and heatmap-based localization, the project aims to

demonstrate how computational imaging techniques can improve both the performance and interpretability of dermatology AI systems.

Milestones

Week 1 (11/11 - 11/17): Literature Review and Project Planning

- Complete literature review on melanoma classification, skin-tone diversity, and dermatology AI bias [1]-[4].
- Review existing melanoma classification models and select a pretrained baseline model (e.g., EfficientNet, ResNet, or DenseNet).
- Identify datasets and determine training, validation, and testing strategy.
- Finalize software architecture and evaluation metrics.

Week 2 (11/18 - 11/24): Data Collection and Baseline Pipeline

- Acquire and preprocess the datasets.
- Implement the baseline melanoma classification workflow using the selected pretrained model.
- Evaluate baseline performance on different demographic or skin-tone subgroups.

Week 3 (11/25 - 12/1): Image Augmentation Development

- Design and implement skin-tone-aware image augmentation techniques.
- Generate augmented training data.
- Retrain or fine-tune the baseline model using augmented datasets.

Week 4 (12/2 - 12/8): Model Evaluation and Analysis

- Compare baseline and augmented model performance.
- Evaluate accuracy, precision, recall, F1-score, and confusion matrices.
- Develop explainability visualizations (e.g., Grad-CAM heatmaps).

Week 5 (12/9 - 12/16): Results, Documentation, and Final Report

- Finalize experimental results and performance comparisons.
- Finalize the web application interface.
- Prepare figures, tables, and visualizations.
- Submit final code and project deliverables on **12/16**.

References

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